

KONSTANTINOS MARONIKOLAKIS

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ACADEMIC POSITIONS

Bilkent University
Visiting Assistant Professor

September 2024 - March 2026

Bilkent University
Assistant Professor

April 2026 - present

EDUCATION

University College Dublin

2020 - 2024

PhD in Mathematics and Statistics

Thesis Title: Universality and Boundary Behaviour of Holomorphic Functions

Supervisor: Dr Myrto Manolaki

National and Kapodistrian University of Athens

2015 - 2019

BSc in Mathematics

Grade 8.93/10 (Excellent)

RESEARCH INTERESTS

My research lies in Complex Analysis with strong connections to Approximation Theory, Operator Theory, and Potential Theory. I am particularly interested in phenomena of universality and boundary behaviour of holomorphic functions, as well as in the geometry of classical Banach spaces of holomorphic functions such as Hardy, Bergman, and Dirichlet spaces.

Several themes unify my work:

- **Universality phenomena:** I study universal holomorphic functions that exhibit “wild” approximation properties, where simple sequences (like partial sums or dilates) can approximate large classes of functions.
- **Boundary behaviour:** I investigate how holomorphic functions on the unit disk behave as we approach the unit circle, particularly focusing on exceptional behaviour.
- **Approximation:** A main tool throughout my research has been approximation results in different spaces and settings, including incomplete polynomial approximation, quantitative polynomial approximation and approximation in Banach spaces of holomorphic functions.

GRANTS AND ACADEMIC DISTINCTIONS

1. Irish Research Council Government of Ireland Postgraduate Scholarship, 2020 - 2024: *funding for doctoral studies in Ireland, 93,400 €*
2. Irish Research Council Ulysses Award 2022 (with Stéphane Charpentier and Myrto Manolaki): *the aim of the Ulysses scheme is to foster new collaborations between Ireland and France-based researchers by providing seed funding for reciprocal travel visits, 5,000 €*
3. 2024 France Excellence Research Residency grant: *this grant is for international researchers in Ireland who want to do a short mobility in France, from two to four weeks, 1,590 €*
4. Kovalevskaya travel grant, July 2022: *the Kovalevskaya travel grant was established to support graduate students to attend the International Congress of Mathematicians 2022, 300 € (eventually held virtually)*
5. Invitation to attend the RIA Hamilton Masterclass with Professor Terence Tao, Royal Irish Academy, Ireland (held virtually), October 2020

RESEARCH ARTICLES

12. C. Bénéteau, R. Cheng, C. Felder, D. Khavinson, M. Manolaki, and K. Maronikolakis. Metric projections, zeros of optimal polynomial approximants, and some extremal problems in Hardy spaces. *Canad. J. Math.*, available online, see also *arXiv: 2511.08000*, 2026

11. S. Charpentier and K. Maronikolakis. Approximation by strongly incomplete polynomials on compact sets. *Concrete Operators*, 13(1):20250015, 2026
10. S. Charpentier, M. Manolaki, and K. Maronikolakis. Invariance of Abel universality under composition and applications. *Indiana Univ. Math. J.*, 75(1):285–308, 2026
9. K. Maronikolakis. Simultaneous approximation by finite Blaschke products and bounded universal functions. submitted, arXiv: 2511.06543, 2025
8. S. Charpentier and K. Maronikolakis. New results on universal Taylor series via weighted polynomial approximation. submitted, arXiv: 2506.20224, 2025
7. S. Charpentier and K. Maronikolakis. Quantitative incomplete polynomial approximation and frequently universal Taylor series. submitted, arXiv: 2504.20240, 2025
6. S. Charpentier, M. Manolaki, and K. Maronikolakis. Abel universal functions: Boundary behaviour and Taylor polynomials. *Rev. Mat. Iberoam.*, 41(3):867–890, 2025
5. K. Maronikolakis. Universal radial approximation in spaces of analytic functions. *J. Math. Anal. Appl.*, 512(1):126102, 2022
4. G. Gavrilopoulos, K. Maronikolakis, and V. Nestoridis. Universal Taylor series in several variables depending on parameters. *Comput. Methods Funct. Theory*, 22(2):261–275, 2022
3. K. Maronikolakis and G. Stamatiou. Universal power series of Seleznev with parameters in several variables. *Monatsh. Math.*, 195(3):477–488, 2021
2. M. Manolaki, K. Maronikolakis, and V. Nestoridis. One-sided extendability of bounded functions. *Complex Anal. Synerg.*, 7(15), 2021
1. K. Maronikolakis and V. Nestoridis. An extension of the universal power series of Seleznev. *Results Math*, 75(151), 2020

CONFERENCE/SEMINAR TALKS

21. “Metric projections and optimal polynomial approximants”, Bench Math Session, Jagiellonian University, Poland, February 2026 (**invited speaker**)
20. “Simultaneous approximation by finite Blaschke products”, 19th Panhellenic Conference of Mathematical Analysis, National Technical University of Athens, Greece, December 2025 (**invited speaker**)
19. “Projections onto invariant subspaces in Hardy spaces”, Mathematical Analysis Workshop and Popular Science Talks, Deskati, Greece, August 2025 (**invited speaker**)
18. “Incomplete polynomial approximation and frequent universality”, Advanced Courses in Operator Theory and Complex Analysis, Université Clermont Auvergne, France, June 2025
17. “Polynomial approximation and applications to frequent universality”, Math Seminar, Sabanci University, Turkey, March 2025 (**invited speaker**)
16. “Polynomial approximation and frequent universality”, Analysis Seminar, National Technical University of Athens, Greece, March 2025 (**invited speaker**)
15. “Weighted polynomial approximation and applications to frequent universality”, Bench Math Session, Jagiellonian University, Poland, February 2025 (**invited speaker**)
14. “Universal Objects in Complex Analysis”, Analysis seminar, Middle East Technical University, Turkey, November 2024 (**invited speaker**)
13. “Invariance of Abel universality under composition”, Advanced Courses in Operator Theory and Complex Analysis, University of La Laguna, Spain, June 2024
12. “Universality and boundary behaviour of holomorphic functions”, Séminaire Analyse & Géométrie, Aix - Marseille University, France, May 2024 (**invited speaker**)
11. “Holomorphic functions with chaotic radial behaviour”, Analysis Seminar, Indiana University Bloomington, USA, March 2024 (**invited speaker**)
10. “Composition problems of Abel universal functions”, Analysis Winter Meeting, Aristotle University of Thessaloniki, Greece, December 2023 (**invited speaker**)

9. “Abel universal functions: properties and variations”, Functional Analysis Seminar, University of Lille, France, December 2023 (**invited speaker**)
8. “Holomorphic functions with chaotic radial behaviour”, Complex Analysis Seminar - Bench Math Session, Jagiellonian University, Poland, December 2023 (**invited speaker**)
7. “Properties of Abel universal functions”, Universality, Zeta-Functions, and Chaotic Operators Workshop, Centre International de Rencontres Mathématiques, France, August 2023 (**invited speaker**)
6. “Properties of Abel universal functions”, Advanced Courses in Operator Theory and Complex Analysis, Aristotle University of Thessaloniki, Greece, June 2023
5. “Properties of Abel universal functions”, Key Lemmas in Analysis and Dynamics, University College Dublin, Ireland, January 2023 (**invited speaker**)
4. “Properties of Abel universal functions”, Journée d’Analyse Harmonique, Aix - Marseille University, France, December 2022 (**invited speaker**)
3. “Holomorphic functions with universal radial limits”, One Day Function Theory Meeting, De Morgan House, UK, September 2022 (**invited speaker**)
2. “Universal radial approximation in spaces of analytic functions”, Computational Methods and Function Theory, Federico Santa María Technical University, Chile (held virtually), January 2022 (**invited speaker**)
1. “Universal radial approximation in spaces of analytic functions”, ComPlane: the next generation (held virtually), June 2021 (**invited speaker**)

Departmental talks:

11. “Approximation by finite Blaschke products”, Analysis seminar, Bilkent University, Turkey, March 2026
10. “Metric projections in Hardy spaces and optimal polynomial approximants”, Analysis seminar, Bilkent University, Turkey, October 2025
9. “Universality and boundary behaviour of holomorphic functions”, Analysis seminar, Bilkent University, Turkey, November 2024
8. “Universal Objects in Complex Analysis”, Analysis seminar, Bilkent University, Turkey, November 2024
7. “Properties of Abel universal functions III: invariance under composition”, Analysis seminar, University College Dublin, Ireland, November 2023
6. “Properties of Abel universal functions II: boundary behaviour”, Analysis seminar, University College Dublin, Ireland, October 2023
5. “Properties of Abel universal functions”, Analysis seminar, University College Dublin, Ireland, April 2023
4. “Universality in analysis”, Mathematics and Statistics Postgraduate Workshop, University College Dublin, Ireland, May 2022
3. “Universal radial approximation in spaces of analytic functions”, Analysis seminar, University College Dublin, Ireland, April 2022
2. “Universality in analysis”, Postgraduate Seminar Series, University College Dublin, Ireland (held virtually), April 2021
1. “Seleznev universal series”, Complex and Harmonic Analysis Seminar, National and Kapodistrian University of Athens, Greece, May 2019

CONFERENCE/SEMINAR PARTICIPATION

14. Hilbert Function Spaces, University of Rome Tor Vergata, Italy, June 2025
13. Analysis seminar, Bilkent University, Turkey, 2024 - present
12. One Day Function Theory Meeting, De Morgan House, UK, September 2023
11. 34th International Workshop on Operator Theory and its Applications, University of Helsinki, Finland, July 2023
10. Topics in Complex Dynamics, Institute of Mathematics at the Universitat de Barcelona, Spain, June 2023

9. Second Congress of Greek Mathematicians, National Technical University of Athens, Greece, July 2022
8. Workshop in memory of Dimitris Gatzouras, National and Kapodistrian University of Athens, Greece, March 2022
7. Transcendental dynamics and beyond: Topics in Complex Dynamics, Centre de Recerca Matematica (held virtually), April 2021
6. AMS/MAA Joint Mathematics Meetings, Washington, D.C., USA (held virtually), January 2021
5. LMS lectures on Holomorphic Dynamics, London Mathematical Society, UK (held virtually), October - December 2020
4. AMS Fall Southeastern Sectional Meeting, University of Tennessee at Chattanooga, USA (held virtually), October 2020
3. Analysis seminar, University College Dublin, Ireland, 2020 - 2024
2. Complex Analysis video seminars, University College London, UK (held virtually), 2020 - 2024
1. Seventh Summer School in Operator Theory, National and Kapodistrian University of Athens, Greece, July 2018

CONFERENCE/SEMINAR ORGANIZING

2. Complexiko: an online seminar series of Greek mathematicians working in Complex Analysis, 2024, co-organised with A. Beslikas, N. Biehler and A. Kouroupis
1. Mathematics and Statistics Postgraduate Workshop, University College Dublin, Ireland, May 2023, co-organised with B. Barbero, T. Comito, T. P. Menezes, B. O'Sullivan, J. Sheekey and J. Williams

TEACHING

- Lecturer at Bilkent University:
 4. Calculus II (**MATH 102**), Spring 2026
 3. Calculus I (**MATH 101**), Fall 2025
 2. Calculus II (**MATH 102**), Spring 2025
 1. Calculus I (**MATH 101**), Fall 2024
- I presented a week-long course on “Linear Dynamics” at the Nesin Mathematics Village, Turkey, August 2024 as part of their Undergraduate and Graduate Summer Camp. The aim of the course was the introduction to the main definitions, results and tools in the topic and it was for an advanced undergraduate and graduate level.
- Tutor at University College Dublin:
 10. Linear Algebra for Science (**MATH10290**), Spring 2024
 9. Group Theory and Applications (**MST30010**), Fall 2023
 8. Metric Spaces (**MATH30090**), Fall 2023
 7. Linear Algebra for Engineers (**MATH10260**), Spring 2023
 6. Group Theory and Applications (**MST30010**), Fall 2022
 5. Metric Spaces (**MATH30090**), Fall 2022
 4. Linear Algebra for Engineers (**MATH10260**), Spring 2022
 3. Group Theory and Applications (**MST30010**), Fall 2021
 2. Metric Spaces (**MATH30090**), Fall 2021
 1. Linear Algebra in the Mathematical and Physical Sciences (**MATH10340**), Spring 2021
- Private tutoring of high school students in Greece from 2015 to 2020

SUPERVISING

Graduate:

1. Ayşe Verda Bal
Tentative Thesis Title: Hypercyclic Taylor shifts on Hardy-Smirnov spaces
Fall 2025 - now

Undergraduate:

1. Ayşe Verda Bal
Project Title: Hypercyclicity of Adjoint Multipliers in Hardy Spaces
Course: Senior Project II (**MATH 492**), Spring 2025

OUTREACH

- Tutor for Junior Mathematical Enrichment, University College Dublin, Ireland, Fall 2022, Fall 2023 and Spring 2024: *The Junior Mathematical Enrichment program provides advanced mathematics classes for secondary school students. I participated as a tutor where I had to interact with the students during problem-solving sessions and also present material on various topics in geometry, algebra and analysis.*
- Outreach Activity as part of Maths Week 2022, University College Dublin, Ireland, October 2022: *As part of the Maths Week, the Museum of Mathematics from Catalonia visited University College Dublin with mathematics exhibitions and interactive puzzles with the goal of outreach to secondary school students. I volunteered as a presenter during the event.*

REFeree WORK

I have refereed for the following journals:

1. Annali di Matematica Pura ed Applicata
2. Complex Analysis and Operator Theory
3. Journal of Analysis

I am also a reviewer for Mathematical Reviews.

REFERENCES

- **Catherine Bénéteau**
Professor, Department of Mathematics and Statistics, University of South Florida
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- **Stéphane Charpentier**
Associate Professor, Aix-Marseille University, CNRS, I2M
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- **Myrto Manolaki**
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